



Princess Sumaya University for Technology
Communication Engineering Department
Applied Electromagnetics
Worksheet 2

1. Given the following vector:

$$\mathbf{B} = \frac{10}{r} \mathbf{a}_r + r \cos \theta \mathbf{a}_\theta + \mathbf{a}_\phi$$

- Express $\mathbf{B}$ in Cartesian and Cylindrical coordinates.
- Find $\mathbf{B}(-3, 4, 0)$ and $\mathbf{B}(5, \frac{\pi}{2}, -2)$.

2. Two uniform vector fields are given by:

$$\mathbf{E} = -5 \mathbf{a}_\rho + 10 \mathbf{a}_\phi + 3 \mathbf{a}_z$$

$$\mathbf{F} = \mathbf{a}_\rho + 2 \mathbf{a}_\phi - 6 \mathbf{a}_z$$

Calculate:

- $|\mathbf{E} \times \mathbf{F}|$.
 - The vector component of $\mathbf{E}$ at $\mathbf{P}(5, \frac{\pi}{2}, 3)$ parallel to the line $x = 2, z = 3$.
 - The angle $\mathbf{E}$ makes with the surface $z = 3$ at $\mathbf{P}$.
3. Let $\mathbf{H} = -5\rho \sin \phi \mathbf{a}_\rho - \rho z \cos \phi \mathbf{a}_\phi + 3\rho \mathbf{a}_z$. At point $\mathbf{P}(2, 30^\circ, -1)$, find:
- A unit vector along $\mathbf{H}$.
 - The component of $\mathbf{H}$ parallel to $\mathbf{a}_x$.
 - The component of $\mathbf{H}$ normal to $\rho = 2$.
 - The component of $\mathbf{H}$ tangential to $\phi = 30^\circ$.